

INTERCONNECTION AS INVESTMENT

*A Clean Energy and Grid Stability Package
for Data Center Development in ERCOT Territory*

A PROPOSAL

HBS 543: Energy Transitions · Dr. Joe Hoagland · Spring 2026
The University of Tennessee, Knoxville

Executive Summary

This proposal offers data center developers in ERCOT territory a partnership package that solves three problems blocking large-load interconnection in Texas: queue access, energy price exposure, and reliability without diesel. The package bundles a Capacity Impact Fee (CIF) that funds shared grid infrastructure, a twenty-year geothermal Power Purchase Agreement (PPA) that locks energy cost on the majority of campus consumption, and a hydrogen fuel cell subscription that replaces diesel backup with clean, contractually firmed long-duration resilience.

Modeled using the approval of five 100 MW campuses, the all-in twenty-year cost is approximately \$1.24 billion: a one-time \$50 million CIF plus \$57–62 million per year in operating contracts. Levelized, this is roughly \$74/MWh delivered, inclusive of grid stability and forty-eight-hour clean backup. The package is \$5–15 million per year more expensive than wholesale-plus-diesel, but each year of accelerated interconnection captures \$300–500 million in revenue at this campus scale. The proposal is structured as three separate contracts with three independent private counterparties. No federal subsidies required. No mandated fuel type. No single point of failure.

1. The Opportunity

Demand for interconnection in ERCOT far outpaces its ability to build capacity. Developers need reliability, speed, and clean energy credentials. The package aligns both interests through a single coordinated transaction with three private counterparties.

What Data Centers Face Today

- More than 238 GW in ERCOT's large-load queue with over 77 percent from data centers, marking a roughly 300 percent year-over-year increase (Electric Reliability Council of Texas 2026a, 2026b; Utility Dive 2026).
- ERCOT VP of System Planning Kristi Hobbs publicly acknowledged at the December 9, 2025 board meeting that the organization has “outgrown the process” for reviewing large loads.
- Data center moratorium bills introduced in roughly a dozen states. 65% of Americans oppose development in their community, with 77% citing rising electricity costs as the primary reason for opposition (Quinnipiac 2026). Hood County, Texas voted 3–2 against a moratorium in February 2026 after state leadership's opposition; Maine's legislature passed (governor vetoed) a first-in-the-nation statewide moratorium in April 2026 (Texas Tribune 2026a).
- S.B. 6 (2025) curtailment obligations for any load ≥ 75 MW during grid events, with PUCT rulemaking active across Projects 58479, 58480, 58481, 58482, and 58484 (Perkins Coie 2025).

What This Package Delivers

- **Interconnection priority:** 500MW of demand prioritized for CIF-paying interconnection requestors in the first batch of ERCOT's new batch study process for large load interconnection, emerging in Summer 2026, after the CIF has been implemented. The CIF satisfies financial commitment for load demand favored by PUCT rulemaking, and guaranteed inclusion in the next available batch (estimated 6 month cycles) will accelerate wait time by an estimated two or more years for the first 500MW of data center demanders willing to sign to the agreements laid out in the proposal.
- **Cost certainty:** \$37.58/MWh locked for twenty years on sixty-nine percent of consumption, with zero fuel-price correlation.
- **New clean generation:** a 500 MW geothermal plant in East Texas built and operated by a private IPP, sold into ERCOT wholesale.
- S.B. 6 compliance with benefits: hydrogen fuel cell backup complies with the required 10–20% curtailable demand, 100-hour annual cap, 30-minute notice. Wholesale Virtual Power Plant integration of backup capacity and behind-the-meter demand response technologies provides revenue when dispatched.

Why East Texas

- **Independent jurisdiction:** PUCT rulemaking moves in 12–18 months without FERC oversight on intrastate matters.

- **Workforce:** Texas oil-and-gas drilling and turbine operations skills transfer directly to enhanced geothermal.
- **Infrastructure and growth:** Gulf Coast water access, existing 345-kV transmission backbone, Houston metro fiber connectivity, and among the lowest commercial electricity costs in the U.S.

2. The Proposal: Three Contracts, Three Counterparties

The package consists of three separate, independently negotiated commercial agreements. Each has one private counterparty. No agreement is contingent on federal subsidy and no agreement mandates any specific fuel type. A developer who prefers a gas PPA can substitute without affecting the CIF or the hydrogen subscription. The framework is fuel-neutral by construction.

CAPACITY IMPACT FEE Developer → TDU	GEOTHERMAL PPA Developer → Private IPP	H ₂ SUBSCRIPTION Developer → H ₂ Provider
~\$50M <i>one-time at interconnection</i>	\$37.58/MWh <i>20-year locked rate</i>	\$25–28M / yr <i>annual reliability subscription</i>
Covers Transmission reinforcement Interconnection facilities Synchronous condensers System studies	Covers 69% of campus consumption Zero fuel-price exposure IPP raises \$2.5B project finance Bank-financed construction	Covers 20 MW critical load 48 hours per campus Replaces diesel backup Provider funds \$1B capital

2.1 Capacity Impact Fee

The CIF is a one-time, proportional payment at interconnection assessed by the TDU. For a 100 MW campus in a five-campus pooled model, it is approximately \$50 million per campus (\$250 million pooled). It covers transmission reinforcement (~\$35M), interconnection facilities (~\$5M), synchronous condensers (\$12–60M), and system studies (\$5–10M). It does not fund generation, each entity funds what it owns.

The CIF extends an instrument the PUCT has already proposed. Draft rule §25.194 in Project No. 58481 (Public Utility Commission of Texas 2026) requires a non-refundable \$50,000/MW interconnection fee plus 100 percent of direct interconnection costs through a Contribution in Aid of Construction. The CIF ensures the full incremental grid infrastructure cost, including transmission reinforcement and stability equipment, is borne by the cost-causer rather than socialized.

Without the CIF, all loads wait multiple years for ERCOT’s general planning process to organically build capacity. With the CIF, participating developers fund the grid infrastructure their load requires; the capacity constraint is removed for those specific interconnections. The CIF’s financial commitment also functions as a self-selection filter that reduces queue congestion for remaining participants. This mirrors the Memphis/Musk precedent, in which a data center developer brought its own generation and paid for all interconnection infrastructure, effectively bypassing the normal queue process.

2.2 Geothermal PPA

A private IPP raises approximately \$2.75 billion in project finance to build a 500 MW geothermal plant in East Texas, configured as ten binary-cycle units of 50 MW each, deployed in stages—Phase 1 at 50 MW (roughly 10–15 well pairs) to prove reservoir performance in Texas geology, scaling to full 500 MW as subsequent phases are financed against additional PPA tranches from additional pool participants. Data centers do not fund construction; they sign twenty-year PPAs at \$37.58/MWh, committing to purchase their proportional share of output. The IPP takes the signed PPAs to a bank. Five creditworthy customers are locked in for twenty years at approximately \$107 million per year combined, and the bank finances the plant against that contracted revenue. This is the same financing structure used in every corporate renewable PPA. The 45Y clean electricity ITC improves the IPP’s economics but is not required for the project to clear.

The choice between geothermal and gas is a financial decision about hedge quality, not a policy statement. The geothermal PPA has zero fuel-price beta: \$37.58/MWh is fixed for twenty years regardless of gas prices or wholesale volatility. A gas PPA carries Henry Hub-linked commodity exposure. More critically, gas sets the marginal price in ERCOT roughly forty percent of hours. When wholesale prices spike to \$5,000–9,000/MWh as during Winter Storm Uri, a gas PPA rises in tandem because it

is correlated with the wholesale market. The hedge fails precisely when needed. Geothermal stays at \$37.58 when wholesale hits \$5,000. When the hedge is correlated with the risk, it is not a hedge.

Mechanically, the PPA is structured as a contract for differences. The data center buys electricity from the wholesale market through a Retail Electric Provider, just like any other customer. If wholesale clears above \$37.58, the IPP pays the data center the difference; if below, the data center pays the IPP. The net effect: effective cost on hedged volume is always \$37.58/MWh. On the unhedged 31 percent, the data center pays prevailing wholesale rates.

2.3 Hydrogen Subscription

Green hydrogen fuel cells are proposed as a targeted clean resilience pilot for ERCOT data centers, not as a replacement for the grid. The purpose is to give large loads a dispatchable backup layer during curtailment, outage, or tight grid conditions, while avoiding the local pollution, fuel delivery, and permitting risks of diesel backup and using fuel cells because it is more efficient to convert hydrogen's chemical energy directly into electricity than to burn hydrogen to run turbines (DOE, 2024; EIA, 2025). Under this structure, the hydrogen provider owns and operates the regional production, storage, delivery, and power sourcing system, while data centers seeking faster and more credible interconnection use the CIF structure to help fund the campus fuel cell portion of a limited clean resilience asset, with dedicated fuel cells and contracted hydrogen reserves functioning as distributed energy resources during grid stress, creating a practical demand channel for green hydrogen and making the hydrogen supply contract part of the operating model rather than an optional clean energy purchase. During low demand or high renewable output hours, the electrolyzer uses clean electricity from PPAs, low price ERCOT hours, independent wind and solar suppliers, or VPP coordinated surplus generation to split water into hydrogen and oxygen for later use, which fits ERCOT well because wind and solar supplied 36% of its electricity demand in the first 9 months of 2025 (EIA, 2025). It shuts down during tight grid hours, so it does not add pressure when reliability is most valuable. The hydrogen subscription functions as portfolio diversification for backup fuel—during Winter Storm Uri, gas generation dropped approximately 30 GW because wellheads and pipelines froze, demonstrating that gas backup is correlated with grid failure. Stored hydrogen on-site is uncorrelated with gas supply constraints, providing backup availability precisely when gas-dependent alternatives cannot. Centralizing production at a regional hub also preserves campus space and helps the provider manage electricity sourcing, emissions accounting, and documentation under the 45V clean hydrogen tax credit framework (Treasury, 2025; IRS & U.S. Department of the Treasury, 2025). Like renewable PPAs and Texas transmission planning, this limited pilot uses long term contracts and coordinated infrastructure to reduce investment uncertainty, support clean energy deployment, generate operating data for ERCOT, PUCT, and private operators, and create system benefits beyond the individual campus.

The proposed pilot uses a 100 MW electrolyzer and 100 MW fuel cell block to provide 48 hours of clean backup for 20 MW of critical load at each of the 5 data centers, making the system large enough to offer real resilience value but still limited enough to remain a focused pilot. Hydrogen City in South Texas provides a useful scale reference, since it proposes a much larger 2.2 GW electrolyzer powered by solar and onshore wind, while this proposal uses a smaller system designed for campus resilience rather than bulk hydrogen production (ABB, 2024). The total hydrogen system is estimated at about \$1 billion, covering the electrolyzer, fuel cells, storage, compression, controls, interconnection, non-potable water treatment, and site work, so any CIF supported contribution toward the campus fuel cell portion would reduce the provider's unrecovered capital need rather than adding a separate cost layer. These estimates are grounded in DOE's PEM electrolyzer benchmark, Caterpillar and Microsoft's 1.5 MW 48 hour data center fuel cell demonstration, and National Energy Technology Laboratory storage cost estimates, with annual operation and input costs estimated at \$25 million to \$35 million (U.S. Department of Energy, 2024; Caterpillar, 2024; Teel, 2024). The value includes \$25M to \$28M in clean resilience subscription payments from each data center, plus event dispatch, peak reduction, tax benefits, and avoided diesel emissions, generating about \$105M to \$130M in annual net operating value, an 8 to 10 year payback period, and a 20 year NPV of about \$296M to \$753M using a 4% to 5% real discount rate consistent with NREL's treatment of long term contracted renewable infrastructure (NREL, 2024). However, since the base calculation treats the \$1B system cost as fully provider financed, any data center contribution toward the campus fuel cell portion would further improve the provider's effective return beyond this estimate.

Although hydrogen is not the cheapest backup option, it offers a different value proposition than natural gas, diesel, batteries, or nuclear. On a per data center basis, natural gas is the cheapest firm capacity alternative at about \$1.4M to \$2.4M annually under EIA cost assumptions, but it still depends on gas delivery, produces onsite emissions, and does not create clean resilience pilot value, diesel displacement value, or renewable surplus integration. Diesel backup is also relatively cheap at about \$3M to \$5M annually, but it brings exhaust, noise, fuel delivery, air permitting, and health concerns, while a 48 hour battery alternative would cost roughly \$16M to \$36M annually, with additional concerns about critical mineral mining, recycling, disposal, and physical footprint. Nuclear would provide clean firm power, but at about \$25M to \$40M annually, its high upfront capital investment, long development and licensing timeline, and public acceptance challenges do not match a near term campus resilience need (EIA, 2025; EPA, 2025; Ericson et al., 2019; NREL, 2024). The calculation is also conservative because it does not monetize health cost savings from reduced diesel exhaust, which would raise the social benefit of the pilot. Over a 20-year contract, the service agreement can include technology refresh and cost sharing clauses so that improvements in electrolyzer efficiency, fuel cell performance, storage, or controls reduce future service costs for participating data centers rather than leaving all innovation gains with the provider. Thus, the pilot is not a system wide mandate, but a limited project with measurable backup service, shared cost responsibility, and a clear operating structure that gives regulators real data before wider rollout, gives ERCOT emergency demand relief and renewable surplus integration, and gives data centers faster interconnection, long term rate certainty, outage protection, grid stress resilience, a cleaner energy portfolio, operational learning, and stronger community and regulatory acceptance under current public pushback (Texas Legislature 2025).

3. Total Investment and Amortization

Customer economics for a 100 MW campus at 95 percent utilization (~832,200 MWh per year) over a twenty-year planning horizon.

3.1 Annual Operating Cost

Component	Per Campus	Counterparty	Contract	Function
CIF (grid infrastructure)	~\$50M one-time	TDU	One-time	Cost internalization
Geothermal PPA (69% of load)	\$22M / yr	Private IPP	20-yr locked	Price hedge, $\beta = 0$
Grid power (31% wholesale)	\$8.9–10.2M / yr	ERCOT	Wholesale	Residual load at market
H ₂ subscription	\$25–28M / yr	H ₂ provider	Annual	Clean resilience
Syncon O&M	~\$1.4M / yr	DC operator	Annual	Stability + flex
TOTAL ANNUAL OPERATING	\$57–62M / yr			+ \$50M one-time CIF

3.2 Twenty-Year Amortization

Spreading the \$50 million CIF across the twenty-year contract horizon, the package levelizes to roughly \$62 million per year, or approximately \$74 per megawatt-hour delivered — all-in, inclusive of grid stability services and forty-eight-hour clean backup. The CIF contributes only \$3/MWh when spread across the campus’s twenty-year consumption.

Metric	Value	Notes
One-time CIF	\$50,000,000	Paid at interconnection (Year 0)
Annual operating (midpoint)	\$59,450,000 / yr	PPA + grid + H ₂ + syncon O&M
20-year nominal total	~\$1,239,000,000	\$50M CIF + 20 × \$59.45M

Metric	Value	Notes
CIF amortized per year	\$2,500,000 / yr	Straight-line over 20 years
Levelized cost per year	~\$62,000,000 / yr	Total ÷ 20
Levelized cost per MWh	~\$74 / MWh	<i>All-in, inclusive of resilience</i>

3.3 Comparison and Premium Analysis

A no-package campus buys all electricity at volatile wholesale rates (~\$29–33M/yr) and maintains diesel backup (~\$15–25M/yr, including fuel storage, permitting, and O&M), totaling roughly \$44–58M/yr or ~\$1.02 billion over twenty years. The package premium is approximately \$219M over the twenty-year horizon, or ~\$11M/yr.

20-year cost	Proposed package	No-package alternative
Upfront (Year 0)	\$50M (CIF)	—
Annual operating (midpoint)	\$59.45M / yr	\$51M / yr
20-year nominal total	~\$1.24B	~\$1.02B
Premium (20-yr)	+\$219M ~\$11M / yr	

The premium is insurance, not cost. It buys (1) a \$37.58/MWh rate locked for twenty years on 69 percent of consumption, fully uncorrelated with gas-driven wholesale spikes; (2) forty-eight-hour clean backup with no air permit risk; (3) SB 6 compliance on negotiated terms operationally frictionless because hydrogen islanding maintains the critical load; (4) twenty-four-seven matched clean baseload from the geothermal plant for net-zero matching without separate REC procurement; and (5) two or more years of interconnection acceleration. At a \$3–5 billion campus scale, acceleration captures \$300–500 million per year of foregone revenue. Two years of captured revenue is roughly \$600 million, nearly three times the twenty-year package premium.

4. Engineered Systems

Three on-site engineered systems plus one off-site generation asset implement the resilience and supply layers of the package.

4.1 Synchronous Condensers (Campus)

Two to four units per campus provide at least 2,000 MW·s of inertia at the 345-kV substation. A synchronous condenser is a free-spinning electric motor connected at 60 Hz. No fuel, no power output, just rotating mass that physically resists frequency collapse when generators trip. At an ERCOT value of lost load of approximately \$35,000/MWh (Brattle Group and Electric Reliability Council of Texas 2024), a one-hour outage at a 500 MW node costs \$17.5 million; the \$12–60M syncon investment is trivially justified against that exposure. The 175 MVA units also provide dynamic VAR support and contribute approximately 3,600 A of fault current at the point of interconnection, improving relay protection. ERCOT's board endorsed the combined ~\$893 million West Texas synchronous condenser project in December 2023 (Electric Reliability Council of Texas 2023), establishing both the technology and the regulatory precedent.

4.2 Hydrogen Fuel Cell Backup (Campus + Regional Hub)

Section 2.3 covered the commercial structure and project economics. Operationally, the hydrogen system works as the campus resilience layer within the broader package. During normal operations, the regional electrolyzer runs only during low demand or renewable surplus hours and shuts down during tight ERCOT conditions, while each campus keeps fuel cells and a contracted hydrogen reserve available for curtailment, outage, or emergency load reduction. When ERCOT activates the Mandatory Flexible Demand Program, the campus can island 20 MW of critical load on fuel cells, reducing grid demand without interrupting core operations. When the fuel cells are not islanded, they can function as distributed energy resources and may be aggregated into the campus VPP with demand response and other behind the meter assets, providing fast deployable capacity during peak hours. The system delivers 4.8 GWh of backup electricity across the 5 campuses during a 48-hour event and can refill the reserve in about 6 days of off peak electrolyzer operation. Water use is managed at the regional hub using recycled or treated wastewater, keeping the production asset, water treatment, compression, storage, and documentation burden away from the campus.

4.3 Virtual Power Plant and ADER Aggregation (Campus)

Campus-adjacent distributed energy resources such as the hydrogen fuel cell block when not islanded, demand response devices, and behind-the-meter storage are aggregated through a utility-managed DERMS under ERCOT's ADER pilot program. Cost of energy offered through VPP enrollment is roughly \$43/kW-year versus \$99/kW-year for a gas peaker. The aggregation provides capacity counted toward ERCOT resource adequacy, participates in Non-Spin and ECRS markets, and accelerates the interconnection study by mitigating grid impact. Policy changes to allow more aggregation of larger DERs and allow flexible demand as capacity resource is beneficial to grid security, and compensation for dispatchable electricity when called upon is beneficial to VPP participants. System costs are separate from the CIF and optional, though participation in the VPP program would allow faster interconnection due to reduced capacity resource requirements.

4.4 The Geothermal Plant (Off-Site)

The 500 MW plant is built and owned by the private IPP. Plant configuration: ten binary-cycle units of 50 MW each, deployed in stages—Phase 1 at 50 MW (roughly 10–15 well pairs) to prove reservoir performance in Texas geology, scaling to full 500 MW as subsequent phases are financed against additional PPA tranches from additional pool participants, sited across South and East Texas near existing 345-kV transmission corridors and where geothermal wells are less than 182°C at depths of a few kilometers. Technology: binary-cycle plants operate as heat exchangers. Hot geothermal fluids ($T < 182^{\circ}\text{C}$) vaporize a secondary working fluid (isobutane or n-pentane) which spins a turbine that drives a generator to produce electricity. The geothermal fluid stays within sealed piping; no air emissions from the working fluid loop. The capital cost is approximately \$2.75 billion and is eligible for the 45Y clean electricity ITC. The plant has black start capability, dispatchable output (ramping up and down generation) by varying secondary fluid flow rate, and synchronous inertia as a byproduct of generation. Binary-

cycle construction and technology risk is borne by the IPP through project finance; the bank underwrites the construction loan. The data center's only exposure is PPA counterparty risk, mitigated by the twenty-year contract and the bank's underwriting.

4.5 Mandatory Flexible Demand Program (Operational Layer)

SB 6 requires that loads ≥ 75 MW be curtailable during grid emergencies. The MFDP proposed here provides better, negotiated terms than whatever generic curtailment requirements emerge from PUCT's ongoing rulemaking: 10–20 percent of contracted demand is curtailable, with a 100-hour annual cap and 30-minute advance notice. The hydrogen fuel cell system makes compliance frictionless. When ERCOT activates the MFDP, each campus islands its 20 MW of critical load on fuel cells. From ERCOT's perspective, hundreds of megawatts of aggregated demand disappear. From the operator's perspective, nothing changes. Operations continue on fuel cell power.

5. System Resilience

The package operates as a layered defense. Each layer activates where the others cannot.

- **Normal operations:** geothermal plant generates into ERCOT wholesale; campus draws at locked \$37.58/MWh PPA on hedged volume; electrolyzer hub runs off-peak only (net-zero load when grid tight); syncons hold 60 Hz at the substation.
- **Grid disturbance (seconds):** syncons inject kinetic energy, slowing the rate of change of frequency and giving operators 10–20 seconds to respond. Campus stays online. Backup systems do not trip because the syncons absorbed the excursion.
- **Curtailed event (minutes to hours):** campus islands 20 MW critical load onto fuel cells. From ERCOT's view, demand disappears; from the operator's view, nothing changes. MFDP commitment satisfied without curtailing real workload.
- **Extended outage (hours to days):** 48 hours of contracted clean backup for 20 MW per campus ($100 \text{ MW} \times 48 \text{ hr} \times 5 \text{ sites} = 4.8 \text{ GWh}$ delivered). Stored H_2 reserve refills in ~6 days at off-peak electrolyzer operation. Geothermal plant's black start capability supports system recovery if outage extends further.

6. Risk Allocation and Policy Pathway

6.1 Risk Allocation

Each entity bears the risk it is best positioned to price.

- **Grid infrastructure risk:** TDU. Proven technology, PUCT-regulated.
- **Construction and technology risk on the plant** — IPP, via project finance. Bank underwrites construction loan based on resource diligence and IPP capability.
- **PPA counterparty risk:** data center, mitigated by 20-year contract and bank underwriting of the IPP.
- **Hydrogen system risk:** H_2 provider. Positive NPV of \$296–753M and 8–10 year payback create strong performance incentive.
- **Federal incentive risk:** 45Y ITC improves IPP economics but is not required. CIF is self-financing through interconnection charges. H_2 provider NPV is positive without credits.
- **Community and environmental risk:** hydrogen fuel cells emit only water vapor (no air permits, no community opposition); electrolyzer hub uses recycled wastewater (~4M gal/yr across five campuses) sited away from data center campuses; Binary cycle has no air emissions and EGS uses lower volumes and pressures than oil-and-gas hydraulic fracturing, with Railroad Commission oversight.

6.2 Policy Pathway

Existing foundation: SB 3 (2021) established PUCT's authority over reliability investments. SB 6 (2025) established large-load obligations and directed PUCT rulemaking, now active across Projects 58479–58484 (Perkins Coie 2025). Draft rule §25.194 already requires \$50,000/MW interconnection fees and 100 percent CIAC (Public Utility Commission of Texas 2026). The President's Ratepayer Protection Pledge (March 4, 2026) commits seven major AI and hyperscale companies to build, bring, or buy their own energy supply — aligning directly with the structure of this package.

Three regulatory additions are needed: (1) the CIF codified as a PUCT-ordered proportional interconnection condition for loads $\geq 75 \text{ MW}$; (2) inertia adequacy standards at nodes with high inverter-based resource penetration, codifying syncon requirements at 345-kV substations; (3) VPP/ADER rulemaking allowing ERCOT to count distributed energy resources as dispatchable capacity. None requires federal action. PUCT rulemaking moves in 12–18 months. ERCOT has already created a new Interconnection and Grid Analysis division and partnered with McKinsey to streamline the large-load process (Electric Reliability Council of Texas 2025b).

6.3 Implementation Timeline

Phase	Milestones
Year 1	PUCT finalizes CIF rulemaking. Binary cycle resource assessment. IPP begins permitting.
Year 2	CIF collected at interconnection. Transmission reinforcement begins. IPP secures project finance. Binary cycle or EGS drilling campaign starts.
Years 3–4	Plant construction. Syncon installation. H ₂ hub commissioning.
Year 5+	Full commercial operation. PPAs active at \$37.58/MWh. 20-year operating horizon begins.

7. Value Summary

This package converts three separate problems — queue access, energy price exposure, and reliability without diesel — into three separate purchases from three private counterparties. Each purchase is priced against a verifiable counterfactual. Each counterparty bears the risk it can best price. No federal subsidy is required.

\$35,000 per MWh ERCOT Value of Lost Load. 1 hr at 500 MW = \$17.5M exposure that syncons mitigate.	48 hours islanding Contracted H₂ backup for 20 MW critical load. No SLA violation.	24/7 clean baseload Geothermal generation matched to load. Net-zero matching without separate RECs.	Day 1 cost certainty Full price at signing. Locked PPA. Two-year acceleration.
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Growth should finance growth. The data centers creating the infrastructure need should pay for the infrastructure. The CIF internalizes those costs to the cost-causer, eliminating the cross-subsidy from 27 million ratepayers. The PPA delivers locked-rate clean energy. The subscription delivers clean long-duration backup. Policy changes to allow the addition of flexible demand with the appropriate amount of additional generative capacity speeds up interconnection. The fact that these voluntary commitments enable \$3.5 billion in private clean energy investment is a consequence of the transactions, not their purpose. The President’s Ratepayer Protection Pledge calls on AI companies to build, bring, or buy all the energy needed for their data centers. The package proposed here is one operational answer.

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Also, either add this citation or change the in text citation from “Texas Legislature 2025” to the final paper’s existing “Texas Legislature 2025” so the citation and reference list match:

Texas Legislature Online. (2025). *89(R) SB 6 Senate committee report version*.
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Appendix

Hydrogen City energy conversion: 280,000 metric tons H₂ per year = 280,000,000 kg H₂. Lower heating value of hydrogen ≈ 33.3 kWh per kg. 280,000,000 × 33.3 kWh = 9,324,000,000 kWh = 9.3 TWh stored hydrogen.

Hydrogen Pilot cost: The system delivers 4.8 GWh of fuel cell electricity during a 48-hour backup event: 100 MW × 48 hours = 4,800 MWh = 4.8 GWh delivered fuel cell electricity. At roughly 55% fuel cell conversion efficiency, this requires about 8.7 GWh of stored hydrogen energy: 4.8 GWh ÷ 0.55 = 8.7 GWh stored hydrogen energy. Producing the full reserve requires about 14 GWh of clean electricity through electrolysis. This is based on about 55 kWh of electricity per kg H₂ and 33.3 kWh of hydrogen energy per kg, which implies roughly 60% electrolysis conversion on a lower heating value basis: 8.7 GWh ÷ 0.60 = 14.5 GWh clean electricity, rounded to about 14 GWh. The electrolyzer can refill the full reserve in about 6 days: 14 GWh ÷ 100 MW = 140 hours = 5.8 days, rounded to about 6 days. Capital cost estimate: DOE PEM electrolyzer benchmark = about \$2,000/kW, so 100 MW electrolyzer ≈ \$200 million. The \$1 billion total includes the electrolyzer, 100 MW fuel cell block, hydrogen storage, compression, controls, interconnection, non-potable water treatment, and site work.

Hydrogen Annual Net Operating value: The annual gross value is estimated at \$140 million to \$155 million, built from a full service clean resilience value of \$125 million to \$140 million across the 5 data centers, plus event dispatch, ERCOT peak reduction value, scarcity price upside, 45V operating benefit, and avoided carbon value from replacing diesel generation. Event dispatch = 100 MW × 100 hours = 10,000 MWh. 10,000 MWh × \$250/MWh = \$2.5 million. ERCOT peak reduction value = 100,000 kW × \$66.76/kW year = \$6.676 million, rounded to \$6.7 million. Scarcity upside is estimated conservatively at \$2.5 million. 45V operating benefit = 10,000 MWh × \$160/MWh delivered fuel cell electricity = \$1.6 million. Avoided carbon value = roughly 7,000 metric tons CO₂ × \$190/metric ton = \$1.33 million, rounded to \$1.3 million. Total annual value = about \$140 million to \$155 million. Subtract annual O&M = \$25 million to \$35 million. Subtract renewable electricity input cost = \$1 million to \$1.5 million. Net operating value = roughly \$105 million to \$130 million.

Hydrogen Payback period: \$1 billion ÷ \$105 million = 9.5 years. \$1 billion ÷ \$130 million = 7.7 years. Rounded payback = 8 to 10 years.

Hydrogen Net Present Value: Initial capital cost = \$1 billion. Annual net operating value = \$105 million to \$130 million. Contract period = 20 years. Real discount rate = 4% to 5%. NPV range = about \$296 million to \$753 million.

Cost of Hydrogen Alternatives: Natural Gas: 100 MW combustion turbine capital cost = 100,000 kW × \$562/kW = \$56.2 million. Annualized capital recovery at 4% to 5% over 20 years = \$56.2 million × 0.074 to 0.080 = \$4.2 million to \$4.5 million per year. Fuel for 100 hours of dispatch = 100 MW × 100 hours = 10,000 MWh. Assuming heat rate around 10,500 Btu/kWh: 10,000,000 kWh × 10,500 Btu/kWh = 105,000 MMBtu. 105,000 MMBtu × \$3.52/MMBtu = about \$0.37 million. Adding O&M, insurance, pipeline and site related costs, and developer return, the total estimated annual cost = about \$7 million to \$12 million per year. Diesel: For a 100 MW diesel backup system, NREL estimates diesel generator capital cost at about \$800/kW and non-fuel operation and maintenance at about \$35/kW year. Therefore, upfront capital cost is = 100,000 kW × \$800/kW = \$80 million. Using a 20-year recovery period and a 4% to 5% real discount rate, the capital recovery factor is about 0.074 to 0.080, so annualized capital cost is = \$80 million × 0.074 to 0.080 = \$5.9 million to \$6.4 million per year. Annual non-fuel operation and maintenance is 100,000 kW × \$35/kW year = \$3.5 million per year. For 100 hours of annual dispatch, the system produces 100 MW × 100 hours = 10,000 MWh = 10,000,000 kWh. Using a diesel fuel consumption rule of about

0.07 gallon/kWh for large diesel generators, annual fuel use is $10,000,000 \text{ kWh} \times 0.07 \text{ gallon/kWh} = 700,000$ gallons. Using EIA's 2025 average diesel price of about \$3.66/gallon, annual fuel cost is $700,000 \text{ gallons} \times \$3.66/\text{gallon} = \$2.56$ million. So, the base annual cost is \$5.9M to \$6.4M + \$3.5M + \$2.56M = \$11.96M to \$12.46M per year. Adding permitting, emissions controls, site compliance, fuel storage, delivery logistics, testing, insurance, operator return brings the estimated cost to about \$15 million to \$25 million per year. Battery: $100 \text{ MW} \times 48 \text{ hours} = 4.8 \text{ GWh}$ battery system. Per firm, the battery alternative would require $20 \text{ MW} \times 48 \text{ hours} = 960 \text{ MWh}$, or 0.96 GWh. NREL projected utility battery costs are benchmarked around shorter duration systems. Using long duration of annualized estimates, the 48-hour battery is estimated at \$80 million to \$180 million per year. Nuclear: Vogtle Units 3 and 4 benchmarks are about 2.2 GW with more than \$30 billion total cost. Annualized equivalent is estimated at \$125 million to \$200 million per year.